Algorithms and Data Structures

REPORT LAB 10

Name: ILGIN
Surname: CEYLAN
Student ID: 82862

TASK1

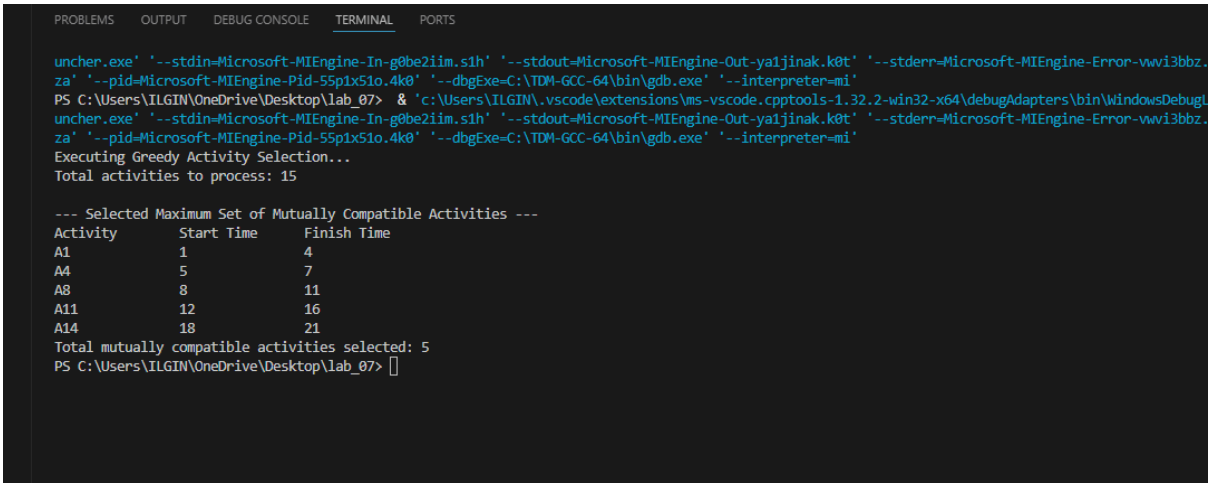
```
String 1: ATCGGCTAATCGATCGGCTAATCGATCGG
String 2: TCGGCTAATCGATCGGCTAATCGATCGGC

LCS Length Table (c)
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
0 1 1 1 1 1 1 1 1 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
0 1 2 2 2 2 2 2 2 2 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
0 1 2 3 3 3 3 3 3 3 3 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4
0 1 2 3 4 4 4 4 4 4 4 4 4 4 4 5 5 5 5 5 5 5 5 5 5 5 5 5 5
0 1 2 3 4 5 5 5 5 5 5 5 5 5 5 5 6 6 6 6 6 6 6 6 6 6 6 6 6
0 1 2 3 4 5 6 6 6 6 6 6 6 6 6 6 6 7 7 7 7 7 7 7 7 7 7 7 7
0 1 2 3 4 5 6 7 7 7 7 7 7 7 7 7 7 7 8 8 8 8 8 8 8 8 8 8 8
0 1 2 3 4 5 6 7 8 8 8 8 8 8 8 8 8 8 8 9 9 9 9 9 9 9 9 9 9
0 1 2 3 4 5 6 7 8 9 9 9 9 9 9 9 9 9 9 9 10 10 10 10 10 10 10 10 10
0 1 2 3 4 5 6 7 8 9 10 10 10 10 10 10 10 10 10 10 10 10 11 11 11 11 11 11 11 11
0 1 2 3 4 5 6 7 8 9 10 11 11 11 11 11 11 11 11 11 11 11 11 12 12 12 12 12 12 12
0 1 2 3 4 5 6 7 8 9 10 11 12 12 12 12 12 12 12 12 12 12 12 12 13 13 13 13 13 13
0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 13 13 13 13 13 13 13 13 13 13 14 14 14 14 14
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 14 14 14 14 14 14 14 14 14 14 15 15 15 15
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 15 15 15 15 15 15 15 15 15 15 16 16 16
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 16 16 16 16 16 16 16 16 16 16 16 17 17
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 17 17 17 17 17 17 17 17 17 17 17 18
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 18 18 18 18 18 18 18 18 18 18 18
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 19 19 19 19 19 19 19 19 19 19
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 20 20 20 20 20 20 20 20 20
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 21 21 21 21 21 21 21 21
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 22 22 22 22 22 22 22
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 23 23 23 23 23 23
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 24 24 24 24 24
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 25 25 25 25
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 26 26 26
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 27 27
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 28
```

```
Direction Table (b) [1:Diag, 2:Up, 3:Left]
2 2 2 2 2 2 1 1 3 3 3 1 3 3 3 3 3 1 1 3 3 3 1 3 3 3 3 3
1 3 3 3 3 1 2 2 1 3 3 3 1 3 3 3 3 1 3 3 1 3 3 3 1 3 3 3 3
2 1 3 3 1 3 3 3 2 1 3 3 3 1 3 3 1 3 3 3 3 1 3 3 3 1 3 3 1
2 2 1 1 3 3 3 3 3 2 1 3 3 3 1 1 3 3 3 3 3 3 1 3 3 3 1 1 3
2 2 1 1 3 3 3 3 3 3 3 1 2 2 2 2 1 1 3 3 3 3 3 3 1 3 3 3 1 1 3
2 1 2 2 1 3 3 3 3 1 3 3 3 1 2 2 1 3 3 3 3 3 1 3 3 3 1 3 3 1
1 2 2 2 2 2 1 3 3 1 3 3 3 3 1 3 3 3 2 1 3 3 1 3 3 3 1 3 3 3 3
2 2 2 2 2 2 2 1 1 3 3 3 1 3 3 3 3 3 3 2 1 1 3 3 3 3 1 3 3 3 3
2 2 2 2 2 2 2 1 1 3 3 3 1 3 3 3 3 3 3 3 1 1 3 3 3 1 3 3 3 3 3
1 2 2 2 2 2 1 2 2 1 3 3 3 1 3 3 3 3 1 3 2 1 3 3 3 1 3 3 3 3 3
2 1 2 2 1 2 2 2 2 1 3 3 3 1 3 3 1 3 3 3 2 1 3 3 3 1 3 3 3 1
2 2 1 1 2 2 2 2 2 2 2 1 3 3 3 1 1 3 3 3 3 3 2 1 3 3 3 1 1 3
2 2 2 2 2 2 2 1 1 2 2 2 2 1 3 3 3 3 3 3 3 1 1 3 3 2 1 3 3 3 3
1 2 2 2 2 2 1 2 2 1 2 2 2 2 1 3 3 3 3 1 3 3 1 3 3 2 1 3 3 3 3
2 1 2 2 1 2 2 2 2 2 1 2 2 2 2 1 3 3 3 1 3 3 3 3 1 3 3 3 2 1 3 3 1
2 2 1 1 2 2 2 2 2 2 2 1 2 2 2 1 1 3 3 3 3 3 3 1 3 3 2 1 1 3
2 2 1 1 2 2 2 2 2 2 2 1 2 2 2 2 1 1 3 3 3 3 3 3 1 3 3 3 1 1 3
2 1 2 2 1 2 2 2 2 2 1 2 2 2 2 1 2 2 1 3 3 3 3 3 1 3 3 3 1 2 1
1 2 2 2 2 1 2 2 1 2 2 2 1 2 2 2 2 1 3 3 1 3 3 3 1 3 3 3 2
2 2 2 2 2 2 1 1 2 2 2 1 2 2 2 2 2 2 1 1 3 3 3 1 3 3 3 3 3
2 2 2 2 2 2 1 1 2 2 2 1 2 2 2 2 2 2 1 1 3 3 3 1 3 3 3 3 3
1 2 2 2 2 1 2 2 1 2 2 2 2 1 2 2 2 1 3 3 3 1 3 3 3 3 3
2 1 2 2 1 2 2 2 2 2 1 2 2 2 1 2 2 1 2 2 2 2 2 1 3 3 3 1 3 3 1
2 2 1 1 2 2 2 2 2 2 1 2 2 2 1 1 2 2 2 2 2 2 1 3 3 3 1 1 3
2 2 2 2 2 2 1 1 2 2 2 1 2 2 2 2 2 2 1 1 2 2 2 1 3 3 3 3 3
1 2 2 2 2 1 2 2 1 2 2 2 1 2 2 2 2 1 2 2 1 2 2 2 1 3 3 3 3
2 1 2 2 1 2 2 2 2 1 2 2 2 1 2 2 2 2 2 1 2 2 2 1 3 3 3 1
2 2 1 1 2 2 2 2 2 2 1 2 2 2 1 1 2 2 2 2 2 2 1 2 2 2 1 1 3
2 2 1 1 2 2 2 2 2 2 1 2 2 2 1 1 2 2 2 2 2 2 1 2 2 2 1 1 3

LCS: TCGGCTAATCGATCGGCTAATCGATCGG
Total LCS Length: 28
PS C:\Users\ILGIN\OneDrive\Desktop\lab_07>
```

TASK2


PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
uncher.exe' '--stdin=Microsoft-MIEngine-In-g0be2iim.s1h' '--stdout=Microsoft-MIEngine-Out-ya1jinak.k0t' '--stderr=Microsoft-MIEngine-Error-vwvi3bbz.
za' '--pid=Microsoft-MIEngine-Pid-55p1x51o.4k0' '--dbgExe=C:\TDM-GCC-64\bin\gdb.exe' '--interpreter=mi'
PS C:\Users\ILGIN\OneDrive\Desktop\lab_07> & 'c:\Users\ILGIN\.vscode\extensions\ms-vscode.cpptools-1.32.2-win32-x64\debugAdapters\bin\WindowsDebugL
uncher.exe' '--stdin=Microsoft-MIEngine-In-g0be2iim.s1h' '--stdout=Microsoft-MIEngine-Out-ya1jinak.k0t' '--stderr=Microsoft-MIEngine-Error-vwvi3bbz.
za' '--pid=Microsoft-MIEngine-Pid-55p1x51o.4k0' '--dbgExe=C:\TDM-GCC-64\bin\gdb.exe' '--interpreter=mi'
Executing Greedy Activity Selection...
Total activities to process: 15
--- Selected Maximum Set of Mutually Compatible Activities ---
Activity Start Time Finish Time
A1 1 4
A4 5 7
A8 8 11
A11 12 16
A14 18 21
Total mutually compatible activities selected: 5
PS C:\Users\ILGIN\OneDrive\Desktop\lab_07>